Practice Quiz: Reading the Prototype's Signals

Part A: Multiple Choice

1. The difference between observation and inference in prototype testing is: A) Observations are always correct; inferences are always wrong B) Observations are raw sensory data; inferences are conclusions drawn from that data using a generative model C) Observations require instruments; inferences do not D) Observations are qualitative; inferences are quantitative

2. Failure events are considered the highest-value signals in testing because: A) They are the most dramatic B) They produce the largest prediction errors, forcing the most significant model updates C) They always indicate that the invention is fundamentally flawed D) They are the easiest to record

3. A user says "this feels cheap" about a prototype. This qualitative feedback is most valuable for: A) Parameter estimation — updating a specific numerical value B) Model structure evaluation — revealing missing variables like texture, weight, and perceived quality C) Confirming that the prototype works as designed D) Determining the manufacturing cost

4. Instrumentation extends the inventor's perceptual reach by: A) Making the prototype more complex B) Transducing hidden states into observable signals, adding sensory channels to the inventor's Markov blanket C) Replacing the inventor with automated testing D) Eliminating the need for qualitative feedback

5. Behavioral observation is uniquely valuable because: A) Users always do what they say they will do B) It captures the user's implicit generative model in action, revealing expectations that verbal feedback cannot articulate C) It is cheaper than other forms of testing D) It always produces quantitative data

6. The Tacoma Narrows Bridge collapse illustrates: A) That quantitative testing is unnecessary B) How signals can be observed but misperceived when the generative

model lacks the relevant causal mechanism C) That bridges should not be built in windy areas D) That failure events are always unpredictable

7. An inventor who relies solely on quantitative measurements risks: A) Collecting too much data B) Building a precisely calibrated version of the wrong thing — a model that is parametrically accurate but structurally incomplete C) Alienating their test users D) Spending too little on prototyping

Part B: Short Answer

1. You test a new umbrella design in wind conditions. The umbrella inverts at 30 mph winds. Record this event as (a) a raw observation, (b) two possible inferences with different generative models, and (c) what type of model update each inference would require.

2. Explain why IDEO's practice of watching users in naturalistic settings (rather than asking them questions in a lab) produces higher-bandwidth perceptual signals. Connect your answer to the distinction between behavioral observations and qualitative feedback.

3. You are testing a software prototype and have access to click logs, page load times, error rates, and post-task survey responses. Design a signal collection strategy that captures all four signal types from the taxonomy, and explain what each uniquely contributes to your understanding of the prototype's performance.